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A Thesis Proposal on:

“Securing Multi-Agent LLM Pipelines: Detecting Prompt Injection Attacks Using a Middleware Framework”

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1. Introduction

The creation of Large Language Models (LLM)-based autonomous agents changed the way complex computational tasks are done in a new way. AutoGen, LangGraph, and CrewAI frameworks enable the building of multi-agent pipelines and also enable the agents to communicate, delegate, and solve complex problems independently. However, it also has a new security issue of prompt injection attacks, in that, evil instructions are also present in data or messages that an LLM is processing. Whereas prompt injection has been viewed as part of single-agent systems, in multi-agent systems, the risk spreads: a compromised message can propagate over a entire pipeline, and affect downstream agents so that they will act in a way they do not intend to, or that is harmful. This is the so-called indirect prompt injection and it is all the more concerning regarding the application of AI in businesses, autonomous decision-making systems, and AI-controlled vital infrastructure. Current multi-agent pipelines of LLM are grounded on implicit trust model in which all inter-agent messaging is held innocent. It is a fallacy, as the messages can be inoculated with antagonistic instructions by other software, databases, web search results or user inputs. These traditional detection techniques, including the use of keywords, fail to detect the attacks which are semantically obfuscated (e.g. Base64 encoding, role-playing prompts, or in directing instructions smuggling). According to this thesis, the proposed system is a full-fledged system capable of detecting and preventing prompt injections attack in multi-agent LLM pipelines with a lightweight, model-agnostic middleware layer called Prompt Injection Detection Middleware (PIDM). It is to guarantee inter-agent communication safety without suffering significant latency or operation overheads, which protect the implementation of autonomous AI agent systems.

2. Literature Review

Prompt Injection Attacks

Perez and Ribeiro (2022) were the first to characterise prompt injection attacks formally and show that LLMs could be manipulated to disregard previous instructions using adversarial inputs. This was later applied by Greshake et al. (2023) to indirect injection, where bad code is concealed in the external material accessed by the model. Liu et al. (2024) divided prompt injection into direct, indirect, and compositional types, and as Liu et al. emphasize, each of them has specific difficulties related to detection mechanisms.

Multi-agent LLM Pipelines Security.

In classical distributed systems, the subject of security in multi-agent systems (MAS) has been examined with a focus on such concepts as Byzantine fault tolerance and secure consensus (Lamport et al., 1982). Nevertheless, the application of these principles to agents based on LLM has not been explored extensively. AutoGen (Wu et al., 2023) and LangGraph (Chase, 2024) presented an example of programmable pipelines that can be used to collaborate between agents

but do not have a security-focused design. Although Yang et al. (2024) showed the agent hijacking situation, they did not suggest any systematic detection and mitigation measures.

Detection and Mitigation Approaches

Current methods of timely injecting detection are mostly based on input preparation, sturdy instruction-following training, or LLM-as-judger structure (Jain et al., 2023; Wei et al., 2023; Zheng et al., 2023). These algorithms are computationally costly and do not support multi-agent and multi-turn scenarios. A requirement for a lightweight, pipeline-aware detector system that can monitor in real time is evident.

Gaps in Existing Research

Along with considerable achievements, there are still gaps in research:

1. Multi-agent pipeline detection that is cascaded is a topic that has not been covered well.
2. A minimum invasive model-agnostic middleware that is capable of minimizing false positives has not been developed.
3. Standardized data does not offer inter-agent message injection attacks.
4. The existing techniques cannot be used to deal with indirect injection attacks and semantically obfuscated attacks.

3. Problem Statement

The existing orchestration models of multi-agent LLMs suppose a benign trust model of inter-agent messaging. Such an assumption puts pipelines at several risks:

1. Absence of inter-agent message validation and sanitization.
2. Exposure to immediate injection by external means via APIs, databases or user contributions.
3. The study of single-agent prompt injection does not include cascading effects of multi-agent systems.
4. Conventional detection systems cannot be effective in semantically obfuscated attacks.

This thesis will solve those issues and create a middleware layer that will identify and isolate malicious messages on the fly and minimize the risk of cascading failures and pipeline performance.

4. Significance

Multi-agent LLM pipelines are essential in securing AI systems in enterprises and autonomous decision-making, as well as AI-driven infrastructure. A credible, timely injection definition will:

- I. Avoid cascading attacks that compromise a number of agents.
- II. Offer standard assessment and defense of pipelines of multi-agents.
- III. Facilitate the safe operation of autonomous agents in the sensitive or high-stakes environments.
- IV. Open-source tools and datasets can also be donated so as to support research in the field of LLM security.

A model-agnostic and simple to deploy middleware such as PIDM will have a low overhead at minimal operational costs and can detect with high accuracy, suitable to both academic and practice.

5. Research Aim and Objectives

Research Aim

To build a lightweight model-agnostic Prompt Injection Detection Middleware (PIDM) that can detect and prevent prompt injection attacks in the multi-agent LLM orchestration pipelines.

Specific Objectives

1. Design and put in place PIDM to monitor inter-agent messages in real time.
2. Find a complete taxonomy of multi-agent prompt injection attacks.
3. It is necessary to create a labelled training and test dataset of benign and adversarial inter-agent messages.
4. Assess the performance of PIDM on precision, recall, F1-score, FP rate and latency.
5. Check whether PIDM can be transferred to two or more MAS frameworks (AutoGen and LangGraph).
6. Create a formal inter-agent communication security threat model.

6. Expected Contributions

1. Initial universal taxonomy of prompt injection attacks on multi-agent LLM pipelines.
2. Open-source, annotated library of inter-agent messages that are adversarial to use in future studies.
3. PIDM: a model-agnostic, lightweight middleware detection and mitigation over real-time.
4. The empirical assessment of determining baseline detection and operation feasibility.
5. The formal threat model of securing autonomous agent communication.

7. Methodology

Phase 1: Threat Modelling & Attack Taxonomy

1. Analyze existing prompt injection attacks and extend them to multi-agent contexts.
2. Define a formal threat model using STRIDE (Spoofing, Tampering, Elevation of Privilege).
3. Develop a taxonomy of at least six attack categories: direct override, role-play smuggling, tool output injection, indirect chain poisoning, encoded payload injection, and context window flooding.

Phase 2: Dataset Construction

- ✓ Generate a labeled dataset of 2,000–5,000 inter-agent messages using real pipeline traces, synthetic injections, and augmented benign messages.
- ✓ Split the dataset 70/15/15 for train/validation/test and ensure human-reviewed label quality.

Phase 3: PIDM Design & Implementation

Rule-Based Filter (RBF): Fast, deterministic first-pass filter using structural heuristics and semantic similarity thresholds.

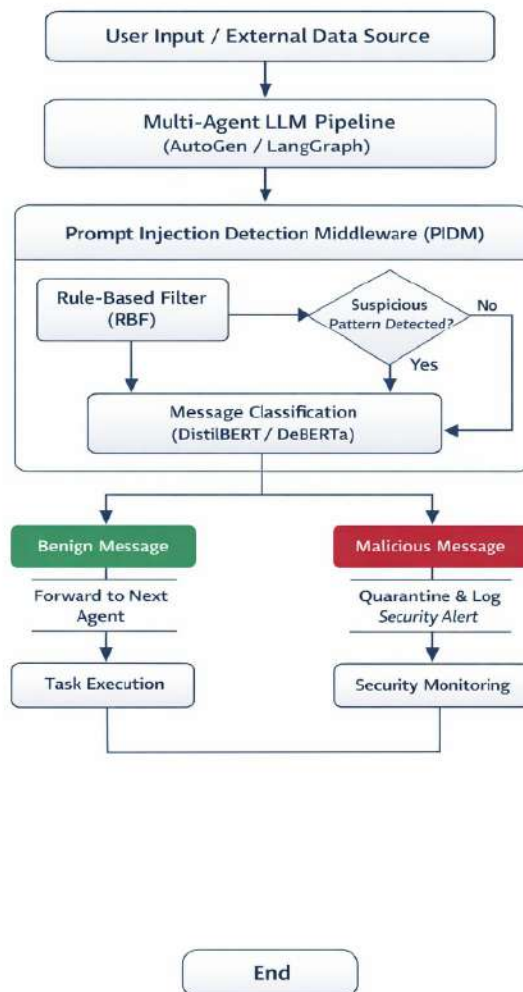
Classifier Layer: Lightweight transformer (DeBERTa-v3-small or DistilBERT) trained on labeled data for binary classification (benign/injected).

Decision & Quarantine Engine: Routes flagged messages to a quarantine queue and logs actions for auditing.

Phase 4: Evaluation & Analysis

- I. Evaluate PIDM on the test set and live pipeline simulation.
- II. Metrics: Precision, Recall, F1-Score, False Positive Rate, Detection Latency, Pipeline Throughput Impact.
- III. Ablation study comparing RBF-only, Classifier-only, and full PIDM.
- IV. Cross-framework validation on AutoGen and LangGraph pipelines.

7.1 Workflow Diagram



8. Technical Stack

MAS Frameworks: AutoGen, LangGraph

LLM APIs: Claude API (Anthropic), GPT-4o (OpenAI)

Detection Model: DeBERTa-v3-small / DistilBERT (HuggingFace)

Programming Language: Python 3.11+

Experiment Tracking: MLflow or Weights & Biases

Dataset Management: Pandas, HuggingFace Datasets

Testing: Pytest, custom pipeline simulation harness

Version Control: Git + GitHub

9. Work Plan and Timeline

Phase	Key Tasks	Timeline	Deliverables
Phase 1	Literature Review & Problem Analysis	Months 1–2	Literature Review
Phase 2	Threat modelling, attack taxonomy	Months 2–3	Threat model document, attack taxonomy report
Phase 3	Dataset construction and labeling	Months 3–6	Labeled dataset (2K–5K messages)
Phase 4	PIDM implementation and testing	Months 6–8	PIDM library v1.0, GitHub repository
Phase 5	Evaluation and ablation study	Months 8–10	Evaluation report, performance metrics
Write-up	Thesis writing and revisions	Months 8–10	Final thesis submission

10. Expected Outcomes

1. PIDM is capable of detecting and quarantining prompt injection attacks in multi-agent LLM pipelines.
2. High detection accuracy with low false positives and minimal latency impact.
3. Open-source dataset and middleware library for future research.
4. Formal threat model guiding secure design of autonomous AI pipelines.
5. Potential scalability for enterprise AI and research applications.

11. Conclusion and Future Work

This thesis aims to develop a robust, lightweight framework for securing multi-agent LLM orchestration pipelines against prompt-injection attacks. The proposed PIDM middleware combines rule-based filtering and transformer-based classification to provide real-time protection while minimizing operational overhead.

Future work may include:

- ✓ Expanding the framework to other LLM orchestration frameworks.
- ✓ Exploring adaptive learning techniques to detect novel injection strategies.
- ✓ Integrating PIDM into production-ready autonomous agent deployments.

References

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